Home / Grade 11 / Algebra / Quarter 2

Control work 4, and the quarter review

The indefinite integral in one paper, and the map that joins approximation, modelling and integration.

LESSON 47–48 | 2 × 40 MIN | ALGEBRA 11, NAZORAT ISHI 4 | P1 · CHAPTER 10 REVIEW

LESSON CLOCK

Instructions	3 min
The paper	42 min
Answers	10 min
Rewrite	20 min
Concept map	10 min
Targets	5 min

LEARNING OBJECTIVES

- Apply every technique of Quarter II in one assessment.
- Rewrite an integrand into integrable form without prompting.
- Build a concept map of the quarter.
- Set a personal target for Quarter III.

TEXTBOOK

National	pp. 221–224
Cambridge	Pure Mathematics 1, pp. 213–216
Workbook	Control paper D

01 Explanation

The paper — 30 marks, 42 minutes

Q	TASK	MARKS	FROM
1	Find $\int (6x^2 - \frac{4}{x^2} + 3\sqrt{x})\, dx$	5	L41–46
2	Find $\int \frac{x^3 - 2x}{x}\, dx$ and $\int (4x - 3)^5\, dx$	5	L41–46
3	Find F with $F'(x) = 3x^2 - 2x$ and $F(2) = 10$	4	L37–40
4	Estimate $\sqrt{63}$ and give the percentage error in a cube's volume from a 0.5% error in its side	5	L28–30
5	$s = t^3 - 12t$: find when the body is at rest and its acceleration then	5	L31–34
6	A cone of height 24 and top radius 8 fills at 6 cm³/s. Find $\frac{dh}{dt}$ at $h = 12$	6	L31–34

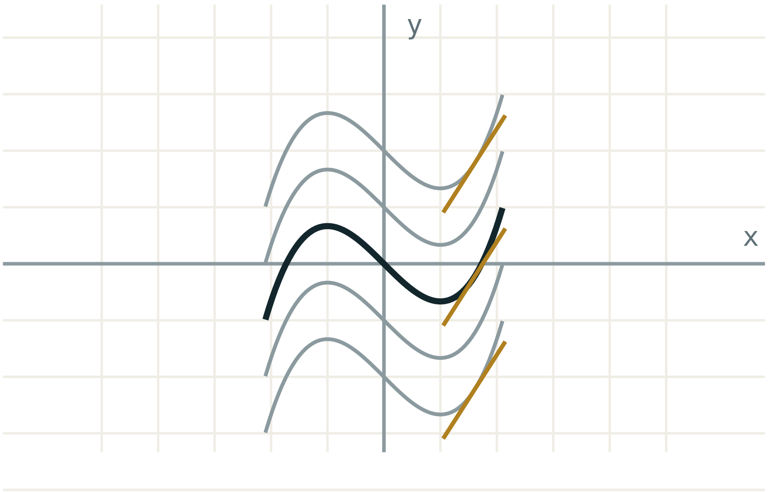
THREE MARKS ARE FOR $+C$ AND UNITS

Q1, Q2 and Q3 each lose a mark without the constant; Q5 and Q6 each lose one without units. Four of the thirty marks are notation.

The concept map

Six boxes, links as sentences:

- **derivative** → **linear approximation** — “near a the tangent is the curve”
- **derivative** → **rate of change** — “with units, in context”
- **rate** → **connected rates** — “the chain rule links two rates”
- **derivative** → **antiderivative** — “run the arrow backwards”
- **antiderivative** → **$+ C$** — “the derivative forgets the height”
- **rules of differentiation** → **rules of integration** — “power up, divide; linearity survives; product does not”



every member has the same derivative — only C differs

The picture behind two of the six boxes.

Looking forward

Quarter III opens with the definite integral and the Newton–Leibniz formula, which turns every antiderivative of this quarter into an area. Nothing new about integration itself is added first — what is added is a pair of limits and a subtraction.

ONE HABIT TO CARRY FORWARD

Rewrite before integrating. Roots into fractional indices, fractions into negative indices, products expanded, quotients divided through. The integral is then one line.

02 Worked examples

EXAMPLE 1 **Model answer, Q1:** $\int \left(6x^2 - \dfrac{4}{x^2} + 3\sqrt{x}\right) dx.$

1. Rewrite: $6x^2 - 4x^{-2} + 3x^{(1/2)}$.

2. $2x^3$

3. $+ \dfrac{4}{x}$

   $-4 \cdot \dfrac{x^{-1}}{-1}$.

4. $+ 2x^{(3/2)} + C$

   $3 \cdot \dfrac{2}{3}$.

ANSWER

$$2x^3 + \frac{4}{x} + 2x\sqrt{x} + C$$

EXAMPLE 2 **Model answer, Q6: cone, height 24, top radius 8, filling at *6* cm³/s.**

1. $r = \dfrac{h}{3}$

   Similar triangles.

2. $V = \dfrac{1}{3}\pi\dfrac{h^2}{9}h = \dfrac{\pi h^3}{27}$

3. $\dfrac{dV}{dh} = \dfrac{\pi h^2}{9} = 16\pi$ at $h = 12$.

4. $\dfrac{dh}{dt} = \dfrac{6}{16\pi} \approx 0.119$

ANSWER

≈ 0.12 cm/s

EXAMPLE 3 **Model answer, Q5: $s = t^3 - 12t$.**

1. $v = 3t^2 - 12 = 3(t - 2)(t + 2)$

2. $t = 2$ (taking $t \geq 0$).

3. $a = 6t = 12$

ANSWER

Rest at $t = 2$ s; $a = 12$ m/s²

03 Interactive model

Work Q1 and Q6 on the board, rewriting the integrand before touching it.

The quarter in ten questions

Estimate $\sqrt{63}$:

- 7.94
- 7.9375
- 7.8
- 8.06

A 0.5% error in a side gives what error in the volume?

- 0.5%
- 1%
- 1.5%
- 3%

$s = t^3 - 12t$: rest at

- $t = 2$
- $t = 4$
- $t = 12$
- $t = 0$

The acceleration there:

- 6
- 12
- 24
- 0

$\int 6x^2\,dx$:

- $12x + C$
- $2x^3 + C$
- $6x^3 + C$
- $3x^3 + C$

$\int \frac{4}{x^2}\,dx$:

- $-\frac{4}{x} + C$
- $\frac{4}{x} + C$
- $4ln|x| + C$
- $-\frac{8}{x^3} + C$

$\int (4x - 3)^5 \, dx$:

- $\dfrac{(4x-3)^6}{6} + C$
- $\dfrac{(4x-3)^6}{24} + C$
- $4(4x-3)^6 + C$
- $(4x-3)^6 + C$

$F' = 3x^2 - 2x$, $F(2) = 10$ gives C:

- 0
- 6
- 10
- -4

$\int \frac{1}{x} \, dx$:

- $\dfrac{x^0}{0}$
- $\ln|x| + C$
- 0
- $x + C$

Cone height 24, radius 8, at 6 cm³/s: $\frac{dh}{dt}$ at $h = 12$

- 0.12
- 0.24
- 1.2
- 0.02

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Concept map	Tushunchalar xaritasi	Карта понятий
Integrable form	Integrallanuvchi shakl	Интегрируемый вид
Constant of integration	Integrallash doimiysi	Постоянная интегрирования
Initial condition	Boshlang‘ich shart	Начальное условие
Self-assessment	O‘z-o‘zini baholash	Самооценка
Target	Maqsad	Цель
Revision	Takrorlash	Повторение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The commonest lost mark in integration is:

- a sign
- $+ C$
- the index
- the units

Before integrating a quotient you should:

- use a quotient rule
- divide through
- square it
- differentiate

Quarter III begins with:

- probability
- the definite integral
- complex numbers
- vectors

A connected-rate problem must first be reduced to:

- two variables
- one variable
- three variables
- a constant

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\int 6x^2\,dx$

   ANSWER $2x^3 + C$

2. $\int \frac{4}{x^2}\,dx$

   ANSWER $-\frac{4}{x} + C$

3. $\int 3\sqrt{x}\,dx$

   ANSWER $2x^{(3/2)} + C$

4. Estimate $\sqrt{63}$

   ANSWER ≈ 7.9375

5. $s = t^3 - 12t$; rest at

   ANSWER $t = 2$

6. Acceleration there

   ANSWER 12

7. $\int (4x - 3)^5\,dx$

   ANSWER $\frac{(4x-3)^6}{24} + C$

Medium · the standard question

7 PROBLEMS

1. $\int (6x^2 - \frac{4}{x^2} + 3\sqrt{x})\, dx$

   ANSWER $2x^3 + \frac{4}{x} + 2x^{(3/2)} + C$

2. $\int \frac{x^3 - 2x}{x}\, dx$

   ANSWER $\frac{x^3}{3} - 2x + C$

3. F with $F' = 3x^2 - 2x$, $F(2) = 10$

   ANSWER $x^3 - x^2 + 6$

4. 0.5% error in a side; error in the volume

   ANSWER 1.5%

5. Cone 24/8 at 6 cm³/s; $\frac{dh}{dt}$ at $h = 12$

   ANSWER ≈ 0.12 cm/s

6. $\int (x + 3)(x - 1)\, dx$

   ANSWER $\frac{x^3}{3} + x^2 - 3x + C$

7. Estimate 4.02^3

   ANSWER ≈ 64.96

Hard · stretch and reason

7 PROBLEMS

1. $\int \frac{(x+2)^2}{x^2}\,dx$

   ANSWER $x + 4\ln|x| - \frac{4}{x} + C$

2. $\int \sqrt{3x+2}\,dx$

   ANSWER $\frac{2(3x+2)^{\wedge}(3/2)}{9} + C$

3. $\int \frac{1}{(2x-5)^3}\,dx$

   ANSWER $-\frac{1}{4(2x-5)^2} + C$

4. Find F with $F'' = 12x$, $F'(1) = 4$, $F(1) = 5$

   ANSWER $2x^3 - 2x + 5$

5. A trough fills at 0.6 m³/min; find $\frac{dh}{dt}$ when the surface is 3 m² in area

   ANSWER 0.2 m/min

6. A car decelerating at 5 m/s² from 30 m/s: stopping distance

   ANSWER 90 m

7. Explain why $\int (x^2+1)^3\,dx$ must be expanded

   ANSWER The bracket rule needs a linear inside

07 Homework

Homework — 4 tasks

Bring the concept map to the first lesson of Quarter III.

1. Rewrite in full every control-work question that lost a mark.
2. Finish the concept map with all six links written as sentences.

3. Find $\int \left(8x^3 - \frac{3}{x^2} + 4\sqrt{x}\right) dx$ and $\int (5x + 2)^4 \, dx$.

4. Write your target for Quarter III in one checkable sentence, and date it.